

DOCUMENT 07

Validation and delivery plan

Acceptance tests, benchmark design and an incremental implementation backlog

1. Test the boundaries before judging the models

All tests below are proposed. No app tests or checkpoint benchmarks were run in this research session.

ID	Test	Required result
T01	Outside-mask writes, each supported engine	Zero changed source samples outside W for a single isolated layer.
T02	Page composition with overlapping layers	Changes are contained in the union of visible applied masks.
T03	Preview/apply identity	Applied support hash equals the accepted preview's support hash.
T04	Padding idempotence: 2 -> 5 -> 2	Final 2-pixel mask equals initial 2-pixel mask exactly.
T05	Zero padding	W equals the selected base footprint; no hidden minimum growth.
T06	Zoom / display scaling	Changing zoom or device-pixel ratio changes no source-coordinate mask bytes.
T07	Feathering	Alpha is zero outside W; confirmed text remains fully covered.
T08	Empty or uncertain base mask	Review/correction outcome; no filled-box fallback.
T09	Fragment and counter preservation	No automatic long bridge or filled artwork gap.
T10	Legacy regression	Same legacy path/results on pinned fixtures; no new default/model requirement.
T11	Missing OCR/script model in all-text policy	No-recognition workflow can start with its actual required models.
T12	Optional SFX model unavailable	Baseline remains usable and the selected fallback is explicit.
T13	Engine swap / retry	Mask geometry does not grow unless explicitly changed and previewed.
T14	Stale preview and cancellation	No old response overwrites a newer plan or cancelled edit.
T15	Padding shrink after applied edit	Old excess pixels are restored from the correct lower composite.

Test oracle

Compare decoded pixel samples, not screenshots, file bytes or perceptual similarity for boundary invariants. Treat image alpha, color depth and source transforms explicitly. For the user's marked references, test geometry intent only; actual mask quality requires an unmarked source.

2. Image quality and persistence cases

ID	Test family	Include
T16	Bubble borders	Text close to curved, jagged and thought-bubble outlines.
T17	Multilingual characters	Japanese, Korean and the other scripts actually intended for release.
T18	Tiny components	Furigana, punctuation, diacritics, detached Hangul parts.
T19	Text appearance	Black, white, colored, outlined, shadowed and low-contrast text.
T20	Difficult geometry	Curved, rotated, broken, very large and partially occluded SFX.
T21	Artwork contact	Letters touching hair, faces, screentone, speed lines and panel borders.
T22	Mask-source disagreement	CTD/other segmenter disagree; review is explicit and no blind broad union occurs.
T23	Round dilation definition	Known circles/corners at each tested radius; no square-growth surprise.
T24	Source edges / strip joins	Negative crop offsets, page edges, variable widths, verified and unverified joins.
T25	Color / depth / export	Supported grayscale/RGB/alpha, 8/16-bit and existing restricted-format behaviors.
T26	Persistence	Save/reopen without optional models; source and mask hashes still match.
T27	Mixed-mode chapters	Legacy and text-shaped patches coexist, export and undo correctly.
T28	Manual masks	Old brush/shape masks stay authoritative and unchanged.
T29	Selection/accessibility	Tiny/disconnected masks, Layers navigation, keyboard actions and click-to-apply.
T30	Resource pressure	Cancellation, out-of-memory, failed provider, model unload and long-strip residency.

Quality review rubric

Score residual lettering, missing outline/shadow removal, damaged non-text pixels, altered balloon/panel borders, reconstructed texture seams and invented marks separately. A mask can pass outside-boundary tests while still being too broad inside its boundary.

Do not claim that exact reconstruction under opaque SFX is objectively known on real pages without a clean reference. Use synthetic overlays on known clean backgrounds for paired reconstruction measurements and real pages for expert usability review.

3. Benchmark design and model promotion

Dataset proposal

Start with a few hundred diverse pages, then expand until rare classes have meaningful samples. Use unmarked, rights-cleared sources. Split by series/book where possible, and keep an untouched holdout outside model/threshold tuning.

Annotate candidate regions, actual lettering/removal pixels, text-owned outlines/shadows, protected artwork and selection groups separately. Keep ambiguous glow/occlusion cases marked as uncertain. Do not turn a text bounding polygon into glyph-mask ground truth.

Ablation sequence

A: current baseline. B: baseline + new mask workflow. C: B + overlapping discovery and CTD line/orphan recovery. D: C + COO specialist. E: best validated discovery + alternative segmenter. F: add PP detector only if needed.

Change one contribution at a time. Keep cleaning policy identical when measuring detector recall; otherwise a gate bypass can be mistaken for a detector improvement.

Metric	Meaning
Instance recall by slice	Fraction of intended text regions found, including SFX and bubble text separately.
False candidates per page	Extra review cost introduced by higher recall.
Complete-page recovery	Fraction of pages where every intended region is discovered.
Text-pixel recall	How much target lettering the base/applied mask covers.
Non-text exposure	Included non-target pixels, with intended padding reported separately.
Protected-art damage	Inclusion/change of annotated artwork or borders.
Correction effort	Human corrections and time to an acceptable page.
Runtime and peak memory	Per page/region on actual target hardware.

A useful exact invariant is $\text{leak_pixels} = \text{count}(\text{changed AND NOT W}) = 0$. For learned quality metrics, do not invent a universal pass threshold. Set targets after baseline measurement and product review; report uncertainty and sample counts by slice.

Promotion gate: improved SFX recovery or mask quality, no unacceptable regression on bubble text/Korean pages, tolerable review cost, device compatibility, and cleared deployment terms. Published Hmean on COO is not a substitute.

4. Implementation backlog and release sequence

Work item	Deliverable	Exit condition
P0 - Pin and diagnose	Record source commit, installed build, model hashes and outcomes on unmarked examples.	Detection, gate, mask and inpainting failures are separated.
P1 - Add geometry policy	legacy/text_shape mode and unit-tested bounded MaskPlan.	T01-T08 and T10 pass on synthetic fixtures.
P2 - Implement refinement	CTD-guided source-resolution base mask and idempotent padding.	Outlines/tiny components reviewed; no rectangular fallback.
P3 - Wire preview and persistence	Versioned mask artifact, region overlay, apply protocol, save/undo/reopen.	T03, T13-T15, T26-T29 pass.
P4 - No-recognition policy	Mode-aware model readiness and explicit all-text/outside-bubble preset.	No OCR/script dependency in that workflow; legacy untouched.
P5 - Improve discovery	Line-map/orphan/tile experiments and candidate fusion.	Incremental gains measured without quality-policy confounding.
P6 - Prototype COO	Exact checkpoint, comparison outputs and multilingual evaluation.	Useful complementary recall; no unverified shipping claim.
P7 - Native optional adapter	Export parity, provider checks, verified model artifacts.	Hardware, security and licensing gates complete.
P8 - Segmenter comparison	Refined CTD vs stroke-model/comic-segmenter experiments.	Add a model only if correction effort or damage improves.
P9 - Release and rollback	Feature-gated rollout, regression pack, user-facing docs.	Old workflow remains available; saved output survives disablement.

Parallel work: P6 research may run while P1-P4 are implemented. Production SFX integration should not bypass the safe-mask contract. P8 is optional, not a reason to block a useful baseline release.

Evidence to attach to the implementation PR

Pinned commit/model manifest; raw unmarked fixtures or authorized fixture references; before/base/padded/output masks; preview/applied hashes; per-slice benchmark counts; device/provider measurements; regression test output; licensing decision record; known failure examples.

Release claim discipline

Say “new optional text-shaped cleaning area” and describe measured coverage gains. Do not claim “all text in all languages,” “perfect SFX reconstruction” or “zero artwork damage” from a small benchmark. Zero writes outside the approved support is a testable boundary claim; correct content inside that support remains an image-quality question.

Source IDs link directly to primary references. Full evidence ledger: Document 08.